NEXT GEN PRECISION FARMING USING ML FOR SUSTAINABLE AGRICULTURAL PRACTICES

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ABSTRACT

In light of current environmental changes, precision agriculture is critical to helping increase crop yield and promoting sustainable farming methods. Traditional farming methods often lack the capacity to adjust to changing conditions such as soil differences and variations in the weather leading to inefficient resource usage. This paper proposes a next-generation precision farming system that integrates IoT-based soil moisture sensors, real-time Weather API's and a TabNet deep learning model used for intelligent crop recommendations. By continuously monitoring the environmental parameters (soils moisture, air temperature, humidity, rainfall, and nutrient levels), the system will use this information to provide accurate and adaptive suggestions of crops to be planted. TabNet will provide high levels of prediction accuracy and will improve model interpretability by allowing features to be assigned attention. By using experimental data, we show that our model will achieve prediction accuracy up to 97.35%, an improvement over other traditional machine-learning methods such as Decision Tree and Random Forest. Additionally, the proposed system will enhance decision-making and will optimise resource use and promote sustainable farming practices.

Keywords: Precision Farming, Sustainable Agriculture, Crop Recommendation, Machine Learning, TabNet, Explainable Artificial Intelligence, Weather API, Agricultural Data Analytics, Deep Learning, Smart Farming

I. INTRODUCTION

The agricultural sector is very important to helping keep the world economy going and ensuring that people have enough to eat. As IoT sensors and weather-based APIs provide more information about our environment, agriculture is becoming increasingly reliant upon data for making decisions regarding how to produce food. Intelligent Crop Recommendation Systems (ICRS) use data to analyze how soil, climate and crop yield interact with each other to be able to make better decisions about how to plant crops and use resources efficiently.

Although these advancements have improved dynamic crop recommendation systems, there are still many barriers to their successful implementation in real-world situations. Many systems perform inconsistently when there are dramatic changes in climate. In addition, many systems rely on outdated methods or limited data integration techniques that do not allow them to take advantage of constant updates of changing input values such as temperature, precipitation, and soil conditions. As a result, these systems are unable to adapt to rapidly changing situations which affects real-time agricultural decision-making.

Gaps present in current systems limited in their current form due to poor interpretability and limited integration of real-time features. Most of the models are not able to provide clarity on the influence of each environmental variable on the model's output, leaving users without enough information to make informed choices about how to implement the model's recommendations. Further, the lack of an efficient method of integrating IoT sensors and weather API data in real time, limits the ability to translate data collected into actionable information. The lack of effective methods of producing multi-dimensional tabular data and poor feature selection practices also limit the ability for these systems to scale, and negatively impacts the overall functionality of the systems.

This project called "Next Gen Precision Farming Utilizing ML for Sustainable Agricultural Practices," the framework will fill these gaps. This project aims to develop a precision farming system using real-time IoT soil sensor data, weather API data, and a TabNet based machine-learning model to address these limitations. The proposed approach is an intelligent system that will process the dynamic environmental interaction by using data to create changing crop recommendations and also have interpretability of the recommendations through a feature selection method that allows for transparency, adaptability, and data-driven decision making to enhance resource use and improve practices of sustainable agriculture.

II. RELATED WORK

The most recent advancement with regard to agricultural technology is the use of AI and advanced technologies, e.g., IoT, machine learning, deep learning, remote sensing networks, and data analysis, in one precision farming platform. Using real-time sensor data from multiple sources (satellite images, predictive analytics, and automated controls) to implement an efficient and next-generation precision farming system will greatly improve agricultural yield, quality of crops, sustainability, and accurate decision-making processes. The greatest challenge to developing a successful smart agricultural system that supplies reliable data regarding many different environmental conditions (e.g., nutrients in the soil, soil chemistry, moisture content, climate conditions, quality of crops, pests, etc.) is that these types of data are very complex and vary on a seasonal basis. An integrated, adaptive, and intelligent agricultural system for the purpose of helping provide food security and to ensure effective utilization of resources in today's society, must be able to monitor, forecast, and optimize agricultural practices on an ongoing basis.

Related Work 1: Smart Agriculture System Using IoT and Machine Learning

[1] This paper describes an agricultural system that combines IoT technology and machine learning to help increase crop yields and automate farming management activities. The system uses Arduino microcontrollers to connect sensors that measure soil moisture, NPK levels, and flame detection (for fire detection). The soil moisture sensor turns on and off the water pump automatically based on preset thresholds for the moisture content of the soil. The NPK sensor provides nutrient information about nitrogen, phosphorus, and potassium and generates recommendations regarding which types of crops or fertilizers are most appropriate for that specific field using a machine learning algorithm (Random Forest). The flame detection sensor provides early detection of fires; this in conjunction with the automatic sprinkler system will help reduce damage from fires.

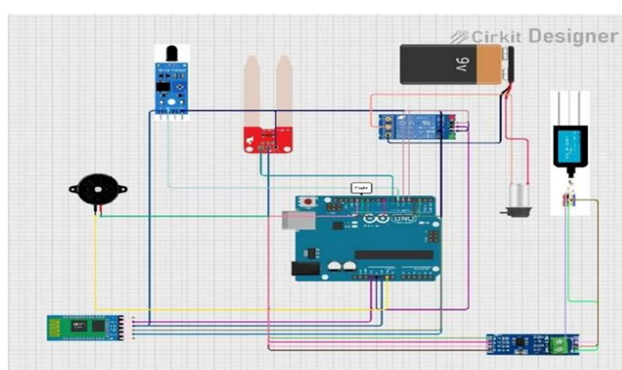

Figure 1: IOT Setup of Smart Agriculture System Using IoT and Machine Learning

The authors identified that real-time monitoring of farm operations, coupled with data-driven decision-making processes, will positively impact farm productivity. The parallel application of IoT devices in conjunction with machine learning will reduce the amount of manual labor associated with implementing farm management practices, optimize the use of fertilizers, prevent excessive irrigation, and maximize crop production. Nonetheless, the study focused only on field-level monitoring; there was no mention of integrating satellite imagery or developing a predictive modeling technique (time-series forecasting) for the farming business model, thus providing opportunities for the development of next generation advanced precision agriculture systems.

Related Work 2: Smart farming using Machine Learning and Deep Learning techniques

[2] This Project describes how precision agriculture uses modern technology (internet-of-things devices, data analytics and machine learning) to increase both the productivity and efficiency of farms. Agriculture is vital for both food production and continued economic growth, yet farmers face many obstacles to successfully growing crops, including an unpredictable climate, poor soil condition and unpredictable crop failure. This study aims to present an alternative to solving these issues through the application of smart farming methods, which involve the gathering and analyzing of data about the physical environment (including soil moisture levels, wind speed, and temperature) in order to improve the quality of the decisions made by farmers. Using IoT devices and machine-driven learning allows farmers to decrease the amount of time spent doing manual labour, maximize how they use their farming resources, and plan for future crops more effectively than they are currently doing.

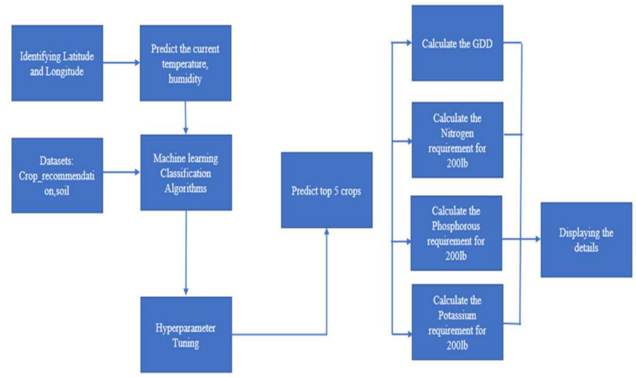

Figure 2: Architecture of Smart farming using Machine Learning and Deep Learning techniques

The Agricultural data collection takes place via multiple sources, such as IoT sensors, historical records, and external factors (e.g., weather). The parameters that will be gathered include soil nutrients (NPK), pH, temperature, humidity, and precipitation. After being collected, the data must undergo preprocessing in order to remove inconsistencies and prepare it for analysis. This involves using machine learning algorithms to train on the processed data so that they may perform some predictive analysis on the collected crop-related data (i.e., recommend which crops to grow based on what is currently being grown, how much of a harvest will come from them, etc.). For example, if a user were to upload an image of weeds or insects through the use of a web-based platform, then [deep] learning algorithms would utilize image processing methods to determine what kinds of weeds/insects are present in the image and provide recommendations concerning what should be used to treat those weeds/insects. Additionally, regression techniques are also performed in order to predict future cultivation costs. As a whole, this includes the entire chain of data collection (from the various sources), preprocessing, training models, and making predictions based on the processed data to provide farmers with an intelligent decision-support tool.

Related Work 3: IoT Based Smart Agriculture Monitoring System and Crop Prediction

[3] This research recommends a method for implementing a holistic approach to precision

agriculture using the Internet of Things (IoT), machine learning for prediction of crop yields, and fuzzy logic to control irrigation with the help of satellite images for creating a correct soil map and predicting yield. The NDVI (Normalized Difference vegetation Index) and NDSI (Normalized Difference Soil Index) were required to identify soil salinity and estimate the health vegetation in an attempt to use the involved Landsat 8 satellite image. Land segmentation was achieved through K-means clustering, while Linear Regression and Random Forest were used to calculate crop yield. Another predictive analytics tool – LSTM (long short-term memory) models were used to predict the future change to the amount of vegetation cover over time.

The data collected from the IoT sensors placed in the soil, including NPK, pH, temperature, and humidity, were analyzed to give farm managers accurate crop recommendations, with 97.35% prediction achieved through Random Forest. Thanks to solar power being employed for operating the irrigation system based on fuzzy logic, there are about 61% savings in the proposed irrigation measures versus the conventional irrigation measures.

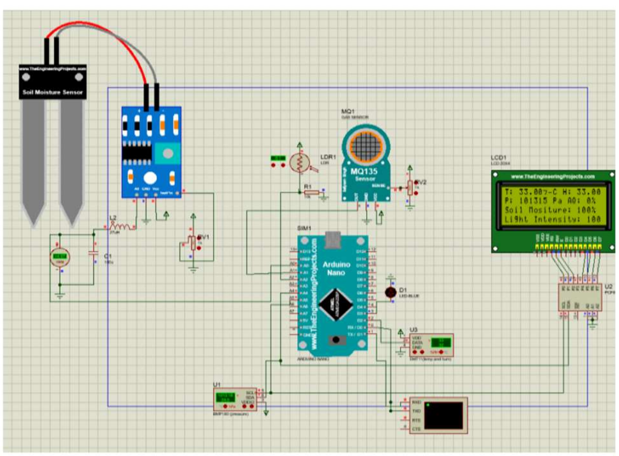

Figure 3: Architecture of IoT Based Smart Agriculture Monitoring System and Crop Prediction

Reference	**Method**	**Dataset/Application**	**Performance**	**Limitation**
[1] Siddhi Ghadge et al., 2024	IoT + Random Forest + NPK, moisture, flame sensors	Soil NPK + environmental monitoring for crop & fertilizer recommendation	Improved yield and automation (no exact metrics)	No quantitative evaluation; limited scalability; no forecasting
[2] S.K.S. Durai & M.D. Shamili, 2022	ML + DL (SVM, RF, CNN, ResNet152V2, Ensemble)	Kaggle datasets + weather scraping + pest datasets	~95.45% accuracy; multi-module system	High computational complexity; requires large datasets
[3] Dineshwar G et al., 2023	IoT + ML (Random Forest, Decision Tree)	Sensor data (temperature, humidity, soil moisture, light)	High prediction capability (not specified numerically)	No standard dataset; lacks time-series & satellite integration
[4] Kasara Sai Pratyush Reddy et al., 2020	IoT + Decision Tree	Soil moisture, temperature, humidity (irrigation system)	Efficient irrigation prediction; water saving	Limited to irrigation; no crop/disease prediction; no metrics
[5] Kumar et al., 2021	ML (SVM, Decision Tree) for crop recommendation	Soil nutrients (NPK), weather parameters	Accuracy around 90% in crop prediction	Limited features; no real-time IoT integration
[6] Liakos et al., 2020	ML & DL in precision agriculture (survey)	Multiple datasets (soil, weather, crop images)	Shows DL improves prediction accuracy significantly	Survey only; no implementation
[7] Grari et al., 2023	IoT + ML/DL for fire detection & prediction	Environmental data (temperature, humidity, CO, light)	High detection efficiency using DL models	Mostly synthetic datasets; lacks real-world validation
[8] Gosai et al., 2022	IoT-based smart irrigation system	Soil moisture + mobile app monitoring	Efficient irrigation and user-friendly interface	High cost of components; limited crop intelligence

Metrics	Decision Tree	Ramdom forest	LighGBM	CatBoost
Accuracy(%)	89.5	97.35	96.06	96.21
Precision(%)	88.14	97.24	96.18	96.24
Recall(%)	87.90	97.35	96.06	96.21
F1 Score(%)	88.03	96.98	96.05	96.18
RMSE	0.19	0.045	0.049	0.047
Explainability	Medium	High	High	High

Table 1: Performance Metrics of present Models

In summary, this study integrates all components needed for smart agriculture, remote sensing using satellite imagery to measure condition of soil, the internet of things(IoT), machine learning (predictive analytics), and renewable energy sources (solar energy) smart farming comprehensively. The developed smart farming solution is a lot more than an automation based on sensors and also leverages predictive analytics to enhance the sustainability of agricultural production over time.

III. METHODOLOGY

The Next-Gen Precision Agriculture System is a data-driven crop recommendation system

- Inputting of soil parameter information
- Real-time weather data using APIs
- Deep learning model (TabNet)

The proposed system predicts which crop is best suited for your soil and will provide certainty as well as an explanation of why the crop was chosen.

Step 1: Data Preprocessing :

The first step of the system is concerned with preprocessing the data. This ensures the quality of the data. Sometimes, the data collected from sensors may be noisy or may have missing values. This may affect the efficiency of the model. In such cases, data cleaning is done. Also, normalization of data may be necessary so that all the features are within a certain range. This step is important since if the data is not appropriately preprocessed, it may affect the efficiency of the deep learning model.

Step 2: Data Collection (IoT Sensors):

In this step, the sensors are deployed in the field to obtain real-time data regarding the environment. The sensors are designed to obtain real-time data regarding critical environmental factors such as soil moisture, temperature, and humidity. These environmental factors are critical in the development and growth of the crops. The sensors are then connected to an Arduino board. The Arduino board acts as a link between the physical world and the digital world. The data obtained in this case is real-time data regarding the environment.

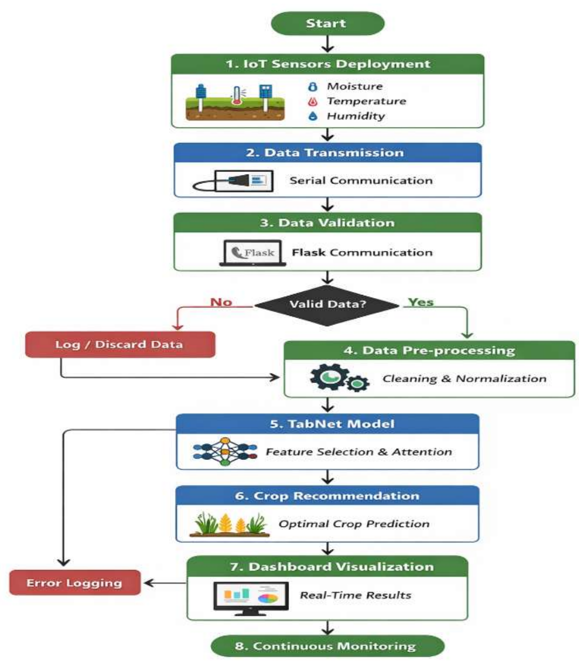

Figure 4: Architecture of Proposed System

Step 3: Data Transmission and Validation :

The data collected will undergo preprocessing prior to being input into the machine learning model to ensure data consistency and compatibility among all features. Categorical variables (soil type and season) will be transformed to numeric values appropriate for the input requirements of the model. Feature scaling will be applied when required to bring numeric values to a normalized and uniform state. In addition to performing missing value validation and aligning features to be consistent with the trained dataset, the final feature vector will be formed by combining the manually entered soil parameters and real-time weather data received from the API.

Step 4:TabNet Based Crop Prediction:

The collected sensor data is then sent to the backend system using serial communication protocol. In this case, the backend server has been implemented using Flask technology to receive the collected sensor data in real time. Validation of the collected sensor data takes place in order to ensure that the collected sensor data is complete, correct, and valid. In case there are any inconsistencies or inaccuracy in the collected sensor data, corrections are made accordingly.

Step 5: Model Processing using TabNet :

After validation, the preprocessed data is fed into the TabNet deep learning model. TabNet is a deep learning model developed specifically for tabular data. TabNet uses a sequential attention mechanism to dynamically select the most relevant features. Unlike traditional models, TabNet does not only provide accurate predictions but also indicates which features contribute most to the prediction. This improves the performance and interpretability of the model. The model processes the input data based on environmental conditions.

Step 6: Crop Recommendation and Output Display:

A model will be trained and will provide the best recommended type of crop based on all of the information that has been collected. The model will format this recommendation in a suitable structured format such as JSON, which will be made accessible to the dashboard through an application programming interface (API)/web interface so that the user or farmer can view present sensor data and the best suitable type of crop for them fast and easily, allowing the user to have an idea of what to do next. The process is ongoing and therefore the model will continue to produce new recommendations each time new information is received; therefore, this system will allow for real-time observation of agricultural conditions.

IV. RESULT ANALYSIS

According to experimental results, the proposed IoT-based precision farming system was able to provide highly accurate crop recommendations through its use of the TabNet model, which effectively captured complex relationships between factors such as soil moisture, temperature, humidity, rainfall, and other seasonal parameters that influence crop production. In addition to providing highly accurate crop recommendations, this system demonstrated solid and consistent performance across many different evaluation metrics when compared with several traditional machine learning models like Decision Trees, Random Forests, LightGBM's and CatBoost models

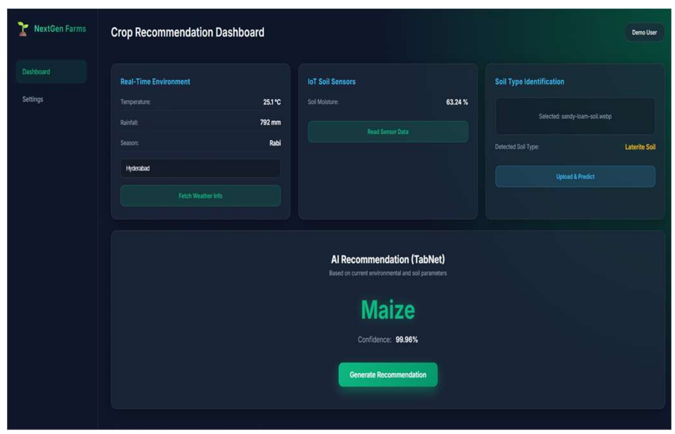

Figure 5: OUTPUT

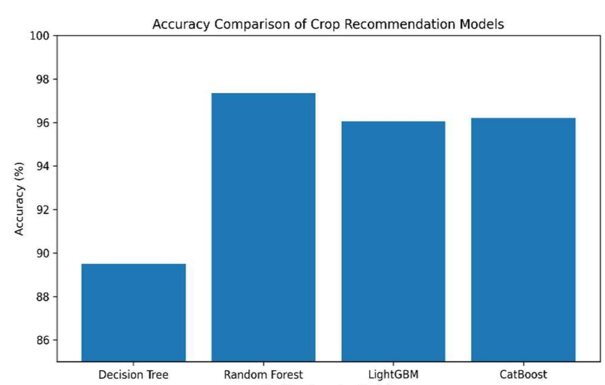

Figure 6: Accuracy and performance comparison of machine learning models for crop recommendation.

According to Fig 6, the accuracy of both advanced ensemble and deep learning approaches exceed that of all the other standard classifiers. The results indicate that the proposed framework has been able to achieve accuracy rates as high as 97.35% and demonstrated strong precision, recall and F1 scores as evidence of stable and well-balanced performance. In addition to high levels of accuracy, the model has also provided clear explanations for the importance of various features that were used in its prediction, and validated

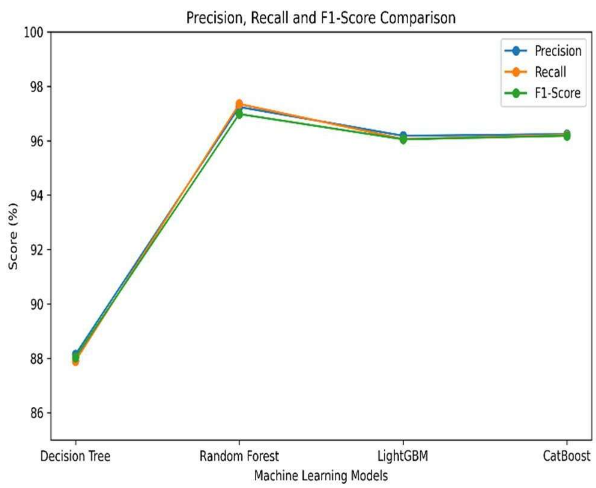

Figure 7 – Precision, Recall & F1-Score Comparison

the significant effects of certain critical agricultural parameters on crop type selection.

V. DISCUSSIONS

The Next Gen Precision Farming System is an innovative system that provides farmers with accurate crop recommendations based on a combination of manual soil parameter inputs and real-time weather information. The use of a web interface allows farmers to easily enter soil moisture, NPK values, pH, soil type, season and location; thereby ensuring that they can access the system, which can then automatically collect temperature, humidity and rainfall information from a weather API.

The integration of this dynamic data into the model provides farmers with current environmental conditions to make recommendations on products to grow, increasing the practical application of the model in real-world agriculture.

The implementation of the TabNet deep learning model improves the reliability and performance of the system. TabNet is designed for use with tabular datasets and has a sequential attention mechanism which determines the most important features that affect prediction results. In addition to accurate crop recommendations, the model also provides a confidence value for each recommendation and provides an explanation of which features contributed most to the prediction, adding transparency to decision-making.

The consistency of the output throughout the workflow, from data preprocessing through to visualising the results of a predictive analysis (TA) on an example farm, assists farmers in making more informed decisions about agricultural planning, while promoting environmentally sustainable farming practices..

VI. FUTURE WORK

In future stages of project development, the priority will be given to increased levels of accuracy and intelligence in system development using current computer vision or machine learning techniques. For example, instead of only using an existing image to analyse soil samples, you could use a deep learning model (e.g., Convolutional Neural Nets - CNN), to analyse what is in the soil and determine whether or not there are any potential diseases affecting that crop from the uploaded image. This will allow for more accurate results to be delivered. Furthermore, the accuracy of crop recommendations will be improved through improved accuracy of the TabNet model with richer datasets containing more attributes associated with soil nutrients (NPK) and pH level, which will in turn enhance the overall quality of decision making for farmers.

At the hardware level, you can easily upgrade your system by adding additional sensors, such as NPK sensors, pH sensors and humidity sensors to gather very localized—and real-time!—data on-site. You can also swap out traditional wired connections for wireless technologies (such as ESP32 or LoRaWAN) that will allow your system to be deployed flexibly over large areas of land. You could even move towards automated irrigation by connecting the system to pumps/valves so you could automatically water your crops based on soil conditions without needing to perform manual tasks.

From the standpoint of usability and deployment, additional capabilities may be added to the system, such as user authentication, data storage and historical analysis to create custom dashboards for each user. Additionally, developing a fertilizer recommendation system, offering multi-language options, using cloud-based platforms to enhance accessibility and usability, as well as developing a mobile app that enables users to use the solution in "real world" situations, will all contribute to achieving scalable deployment of the solution and enable farmers to effectively use the solution in their operations.

VII. REFERENCES

[1] S. Ghage, Y. Tekawade, H. Mutekar, A. Shinde, and S. Gawade,"Smart Agriculture System Using IoT and Machine Learning,"*Int.J.innovative Research in Technology(IJIRT),*vol. 10, no. 2160-2162, 2024.

[2] S.K.SDurai and M.D.Shamili,“Smart Farming Using Machine Learning and Deep Learning Techniques,” *Decision Analytics Journal*, vol. 3, p. 100041, 2022

[3] Dineshwar.G,Nithish p and Jidesh T.,“IoT Based Smart Agriculture Monitoring System and Crop Prediction,” *Int. J. Creative Research Thoughts (IJCRT)*, vol. 11, no. 9, pp. f126–f131, 2023.

[4] K. S. P. Reddy, Y. Mohana Roopa, K. R. L. Narasimha, and N. S.Nandan, “IoT Based Smart Agriculture Using Machine Learning,” in *Proc. 2nd Int. Conf. Inventive Research in Computing Applications (ICIRCA)*, 2020, pp. 131–135.

[5] O. Friha, M. A. Ferrag, L. Shu, L. Maglaras, and X. Wang, “Internet of Things for the Future of Smart Agriculture: A Comprehensive Survey,” *IEEE/CAA J. Automatica Sinica*, vol. 8, no. 4, pp. 718–752, 2021.

[6] G. Liakos, P. Busato, D. Moshou, S. Pearson, and D. Bochtis, “Machine Learning in Agriculture: A Review,” *Sensors*, vol. 20, no. 5, pp. 1–29, 2020.

[7] A. Kamilaris and F. X. Prenafeta-Boldú, “Deep Learning in Agriculture: A Survey,” Computers and Electronics in Agriculture, vol. 147, pp. 70–90, 2020.

[8] M. A. Ferrag, L. Maglaras, H. Janicke, J. Jiang, and T. J. Cruz, “Authentication Protocols for IoT-Based Smart Agriculture Systems,” *IEEE Access*, vol. 8, pp. 32067–32084, 2020.

[9] P. Sharma, S. K. Sharma, and A. K. Singh, “Smart Agriculture Using IoT and Machine Learning: A Survey,” *Int. J. Advanced Science and Technology*, vol. 29, no. 6, pp. 1–10, 2020.

[10] R.K.Jain and S.Kumar, “Crop Recommendation System Using Machine Learning,” *Int. J. Recent Technology and Engineering*, vol. 9, no. 5, pp. 120–125, 2021.

[11] S. K. Bera, A. Saha, and S. Roy, “Precision Agriculture Using IoT and Machine Learning,” *Procedia Computer Science*, vol. 192, pp. 134–143, 2021.

[12] M. Muhammed, E. Ahvar, and M. Trocan, “AIoT for Smart Agriculture: Architectures and Applications,” *Journal of Network and Computer Applications*, vol. 200, pp. 103–120, 2023.

[13] M.Grari et al.,“IoT and Machine Learning for Forest Fire Detection: A Review,” *IEEE Access*, vol. 11, pp. 45000–45015, 2023.

[14] A. Verma, R. Singh, and P. Gupta, “Crop Yield Prediction Using Machine Learning Techniques,” *Int. J. Agricultural Informatics*, vol. 15, no. 2, pp. 45–55, 2022.

[15] S. R. Patel and K. Mehta, “Deep Learning-Based Smart Farming System Using IoT,” *IEEE Internet of Things Journal*, vol. 11, no. 3, pp. 2100–2110, 2024.